

House Price Prediction using Linear Regression

GitHub-ready Machine Learning Project

1. Project Overview

This project demonstrates a complete machine learning workflow for predicting house prices using Linear Regression. The workflow covers data loading, exploration, cleaning, visualization, train-test splitting, model training, prediction, evaluation, coefficient interpretation, and residual analysis.

2. Project Structure

```
linear-regression-house-price/
├── data/
│   └── house_data.csv
├── notebooks/
│   └── linear_regression.ipynb
├── README.md
├── requirements.txt
└── .gitignore
```

3. Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)
```

4. Load Dataset

```
df = pd.read_csv("../data/house_data.csv")

df.head()
```

5. Understand the Dataset

```
df.shape
df.info()
df.describe()
df.isnull().sum()
df.duplicated().sum()
```

6. Exploratory Data Analysis

```
plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=df,
    x="Area",
    y="Price"
)

plt.title("Area vs House Price")
plt.xlabel("Area")
plt.ylabel("Price")
plt.show()

plt.figure(figsize=(8, 6))

sns.heatmap(
```

```

        df.corr(numeric_only=True),
        annot=True,
        cmap="coolwarm"
    )

plt.title("Correlation Matrix")
plt.show()

```

7. Feature and Target Selection

```

X = df.drop("Price", axis=1)
y = df["Price"]

```

8. Train-Test Split

```

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

```

9. Train Linear Regression Model

```

model = LinearRegression()
model.fit(X_train, y_train)

```

10. Make Predictions

```

y_pred = model.predict(X_test)

y_pred[:10]

```

11. Model Evaluation

```

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("MAE :", mae)
print("MSE :", mse)
print("RMSE:", rmse)
print("R²  :", r2)

```

12. Interpret Model Coefficients

```

coefficients = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": model.coef_
})

coefficients

print("Intercept:", model.intercept_)

```

13. Actual vs Predicted Prices

```

plt.figure(figsize=(8, 6))

plt.scatter(y_test, y_pred)

plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted House Prices")
plt.show()

```

14. Residual Analysis

```

residuals = y_test - y_pred

plt.figure(figsize=(8, 5))

```

```
sns.scatterplot(
    x=y_pred,
    y=residuals
)

plt.axhline(0, linestyle="--")

plt.xlabel("Predicted Price")
plt.ylabel("Residuals")
plt.title("Residual Plot")
plt.show()
```

15. Prediction on a New House

```
new_house = pd.DataFrame({
    "Area": [1500],
    "Bedrooms": [3],
    "Bathrooms": [2],
    "Parking": [1],
    "Age": [5]
})

predicted_price = model.predict(new_house)

print("Predicted House Price:", predicted_price[0])
```

16. Model Equation

The Linear Regression model learns an equation of the form:

$$\text{Price} = b_0 + b_1(\text{Area}) + b_2(\text{Bedrooms}) + b_3(\text{Bathrooms}) + b_4(\text{Parking}) + b_5(\text{Age})$$

17. GitHub README Content

```
# ■ House Price Prediction using Linear Regression

## Overview
This project uses Linear Regression to predict house prices from property-related features.

## Objective
Build an interpretable regression model and evaluate its performance using standard regression metrics.

## Technologies
Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook.

## Workflow
Data Loading → Data Understanding → EDA → Data Cleaning → Train/Test Split → Model Training → Prediction → Evaluation →

## Evaluation Metrics
- MAE
- MSE
- RMSE
- R2 Score

## Future Improvements
- Feature engineering
- Outlier treatment
- Cross-validation
- Ridge and Lasso Regression
- Random Forest comparison
- Gradient Boosting comparison

## Author
Anushka Tyagi
```

18. requirements.txt

```
numpy
pandas
matplotlib
seaborn
scikit-learn
jupyter
```

Important

Run the notebook with your actual dataset and replace the placeholder evaluation values in the README with your real MAE, RMSE, and R² results before publishing the project on GitHub.